

Aedes eegypti in Türkiye, Ancient Remnant or New Introduction, Current Status And Management Options

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Türkiye

- ▮ First detection
 - ▮ in 2015 exact area Eastern Blacksea area
- ▮ First action
 - ▮ in 2016 during three months

WHEN DID IT START?

2015-2016

- Surveillance
- Density study
- Filling data gaps (especially for invasive species)
- Encouraging and supporting local implementation of vector surveillance through a regional approach

With Vectornet ve Turkish Ministry of Health support



T.C. SAĞLIK BAKANLIĞI
HALK SAĞLIĞI GENEL MÜDÜRLÜĞÜ



- We use ovitraps and human landing catch
- BG sentinel and EVS traps
- But not synchronise count or wekkly/monthly based data

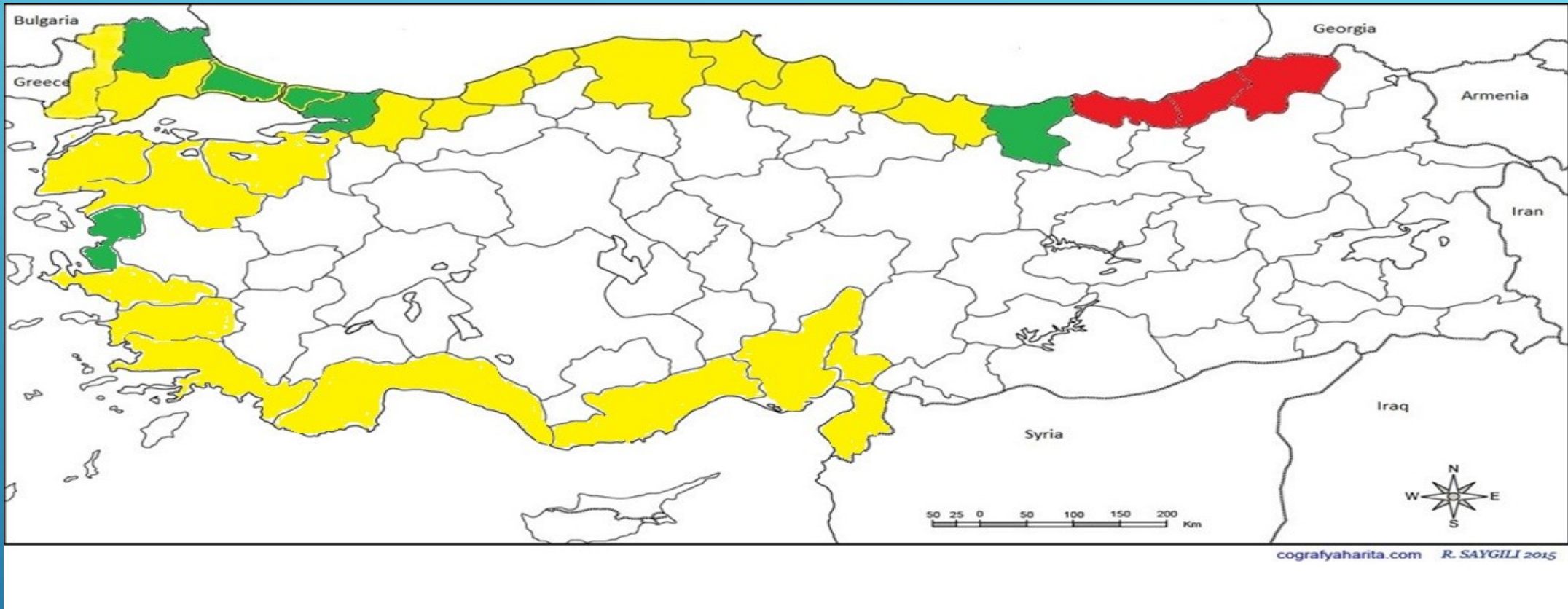


2015-2020

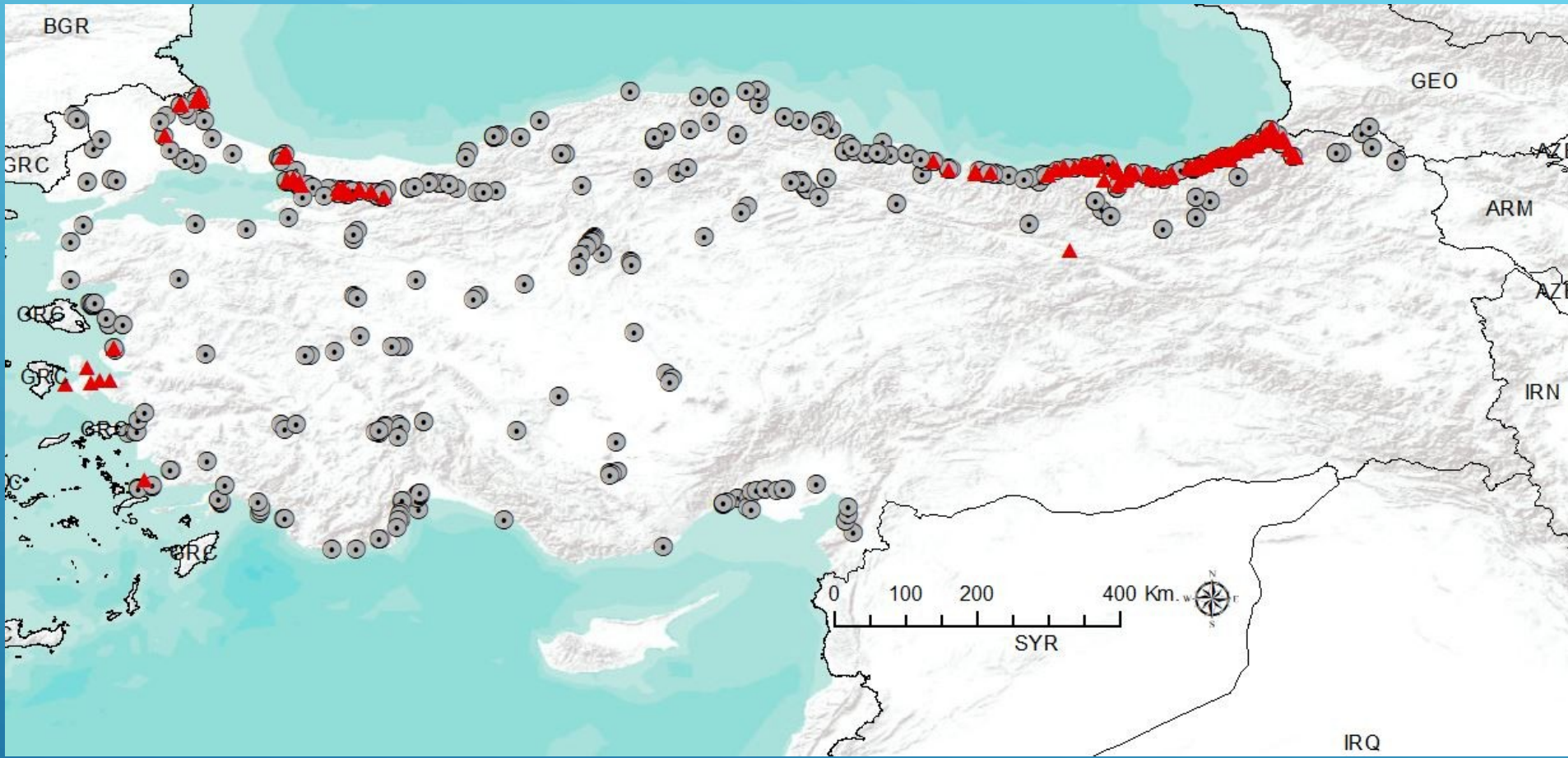








we surveyed all seaside part of Türkiye for invasive *Aedes* species but not found *Aedes aegypti*, found locally established *Ae. albopictus* in 2018-2019



1600 points were checked and 487 points positive with *Aedes albopictus*

New areas

İzmir Urla, Mordoğan, Guzelbahçe

Muğla Yalıkavak

Erzincan

Ordu



700 points were checked for spreading pattern (Artvin, Rize, Trabzon, Giresun, Ordu) 20 points positive with *Aedes aegypti*

Surveillance and Control of Invasive Aedes Species in The Eastern Black Sea Area of Turkey

Türkiye'nin Doğu Karadeniz Bölgesi'nde istilacı Aedes türlerinin izlenmesi ve kontrolü

Muhammet Mustafa AKINER¹, Berna DEMİRCİ², Hilal BEDİR³, Murat ÖZTÜRK⁴, Rıdvan DEMİRTAŞ¹, Ahmet Ferhat DOĞAN⁴, Akgün GÖKDEMİR⁵, Seher TOPLUOĞLU⁶, Ünal ALTUĞ⁵, Zehra Özlem KURTCEBE⁶, Hasan IRMAK⁷

ABSTRACT

Objective: Invasive mosquito species are a huge problem world-wide and can cause serious mosquito borne disease epidemics. Recent surveys in Europe revealed that many autochthonous cases of chikungunya are related to the invasive *Aedes albopictus* and *Aedes aegypti* species. Extensive surveillance of these species and other invasive Aedine mosquito species is required for the prevention and timely response to possible outbreaks. This study focuses on surveillance and control operation success in the Eastern Black Sea region of Turkey.

Methods: This surveillance study was performed during the 2016-2017 vector active season. Three cities were surveyed according to ECDC and CDC guidelines. Control operations were performed during the second half of the 2017 active season under the supervision of the Turkish Ministry of Health's Zoonotic and Vector-borne Diseases Department. IGR, a Bti/Bs mix and liquid Bti were used for larval control. Different pyrethroids, nicotine mimics and natural products were used for

ÖZET

Amaç: İstilacı sivrisinek türleri tüm kıtalarda büyük problem olup birçok alanda sivrisinek kökenli ciddi hastalık salgınlarına neden olabilmektedir. Avrupa Kitasındaki son araştırmalar, pek çok otomom chikungunya vakalarının, istilacı *Aedes albopictus* ve *Aedes aegypti* türleri ile ilişkili olduğunu göstermiştir. Bu türlerin ve diğer istilacı Aedine sivrisinek türlerinin kapsamlı şekilde izlenmesi, muhtemel salgınları önlemek ve zamanında müdahale etmek için gereklidir. Bu çalışma, Doğu Karadeniz bölgesinde izleme ve kontrol çalışmalarının başarısı üzerine odaklanmıştır.

Yöntem: İzleme çalışması 2016-2017 vektör aktif sezonda ve üç şehirde ECDC ve CDC yönergelerine göre gerçekleştirilmiştir. Kontrol çalışmaları, 2017 vektör aktif sezonun ikinci yarısında T.C. Sağlık Bakanlığı Zoonotik ve Vektörel Hastalıklar Dairesi denetimi altında gerçekleştirilmiştir. Larva kontrolü için IGR, Bti/Bs karışımı ile sıvı Bti kullanılmıştır. Ergin kontrolü için farklı pyrethroidler, nikotin taklit eden maddeler ve doğal ürünler kullanılmıştır. Popülasyon dalgalanmaları

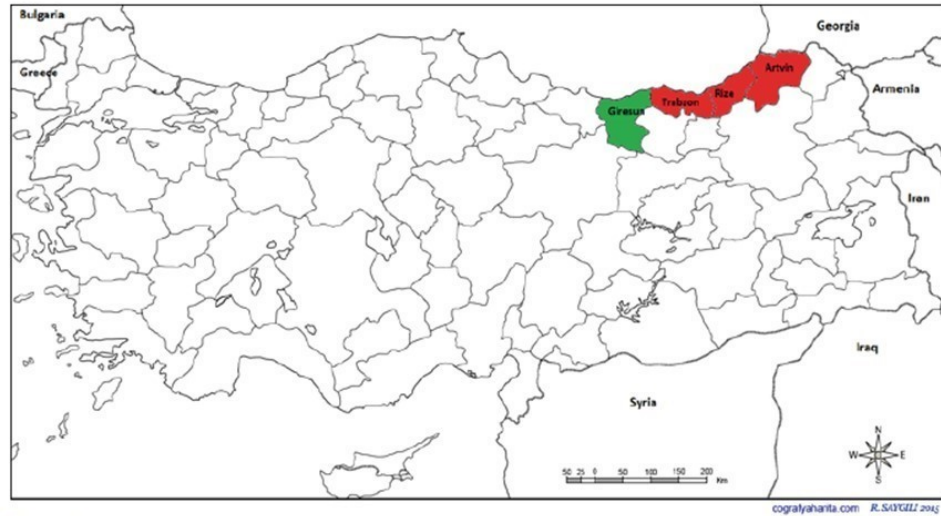


Figure 1. Study and control operations areas during the 2016-17 active vector seasons



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Original article (Orijinal araştırma)

Status of the invasive mosquito species *Aedes aegypti* (L., 1762) and *Aedes albopictus* (Skuse, 1895) (Diptera: Culicidae) in Turkey¹

İşgalci sivrisinek türleri *Aedes aegypti* (L., 1762) ve *Aedes albopictus* (Skuse, 1895) (Diptera: Culicidae)'un Türkiye'deki durumları

Berna DEMİRCİ^{2*}, Hilal BEDİR^{3*}, Murat ÖZTÜRK^{4*}, M. Mustafa AKINER^{4*}

Abstract

Aedes aegypti (L., 1762) and *Aedes albopictus* (Skuse, 1895) (Diptera: Culicidae) are important vectors of arboviruses. In Turkey, *Ae. albopictus* eggs were detected in the Thrace area of northwestern Turkey for the first time in 2011. In 2015, studies revealed the spread of *Ae. albopictus* and the first detection of *Ae. aegypti* within northeastern Turkey was reported. This paper reports the results of a survey of the presence and distribution of *Ae. albopictus* and *Ae. aegypti* in Turkey conducted over 5 years. As of 2019, monitoring studies were conducted on the presence of *Ae. albopictus* and *Ae. aegypti* in five geographical regions (Black Sea, Central Anatolia, Marmara, Mediterranean and Aegean Regions). A comprehensive range of potential larval habitats, such as tires, artificial containers, cemeteries, water bottles and natural breeding habitats, were assessed. In addition, standard ovitraps and adults' traps were used in some localities. This study showed that *Ae. albopictus*, in particular, expanded its distribution each year and has the potential to extend its range throughout Turkey over the next few years. In Turkey, the distribution of *Ae. aegypti* is currently limited to northeastern Turkey. Future work focused on determining more effective surveillance and control studies is discussed.

Keywords: *Aedes aegypti*, *Aedes albopictus*, breeding habitats, mosquito ecology, vector control

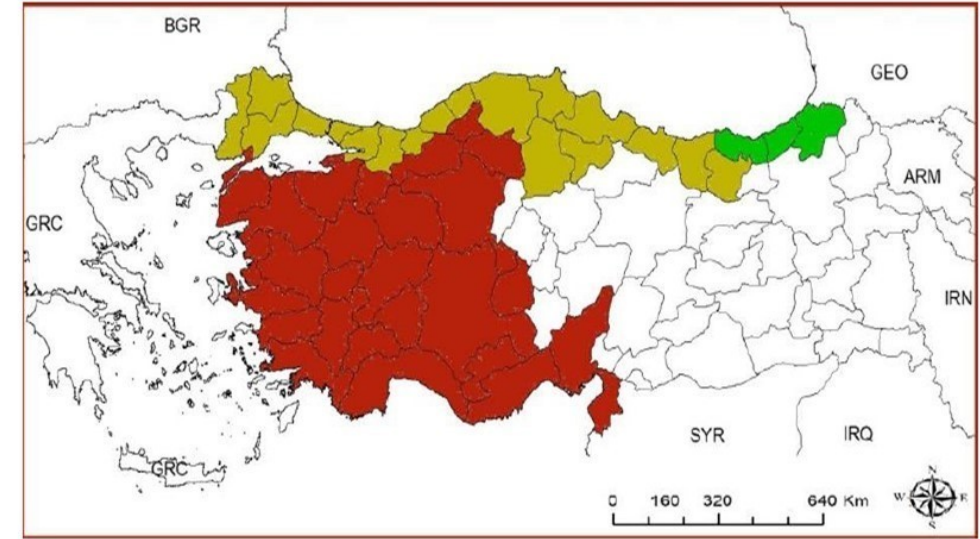


Figure 1. 2015-2019 surveillance area. Green: 2015 surveillance area (East Black Sea Region); Yellow: 2015-2016-2017 surveillance area (West Black Sea and Marmara Regions); Red: 2015-2016-2017-2018 and 2019 surveillance area: (Aegean, Mediterranean and Central Anatolia Regions). ARM: Armenia; BGR: Bulgaria; GE: Georgia; GRC: Greece; IRQ: Iraq; IRN: Iran.



Figure 3. Distribution of *Aedes albopictus* and *Aedes aegypti* in northeastern Turkey in 2015. Blue triangles; monitored sampling sites, yellow circles; positive sampling sites for *Ae. albopictus* and green circles; positive sampling sites for *Ae. aegypti*.



Figure 4. Distribution of *Aedes albopictus* and *Aedes aegypti* in northeastern Turkey in 2016. Blue triangles; monitored sampling sites, yellow circles; positive sampling sites for *Ae. albopictus* and green circles; positive sampling sites for *Ae. aegypti*.

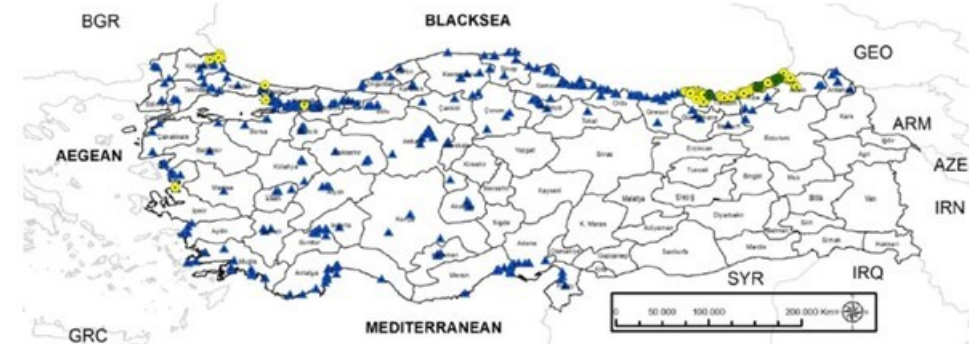


Figure 6. Distribution of *Aedes albopictus* and *Aedes aegypti* in northeastern Turkey in 2018. Blue triangles; monitored sampling sites, yellow circles; positive sampling sites for *Ae. albopictus* and green circles; positive sampling sites for *Ae. aegypti*.

Monitoring studies in five areas - Black Sea, Central Anatolia, Marmara, Mediterranean and Aegean Regions)



Figure 5. Distribution of *Aedes albopictus* and *Aedes aegypti* in northeastern Turkey in 2017. Blue triangles; monitored sampling sites, yellow circles; positive sampling sites for *Ae. albopictus* and green circles; positive sampling sites for *Ae. aegypti*.

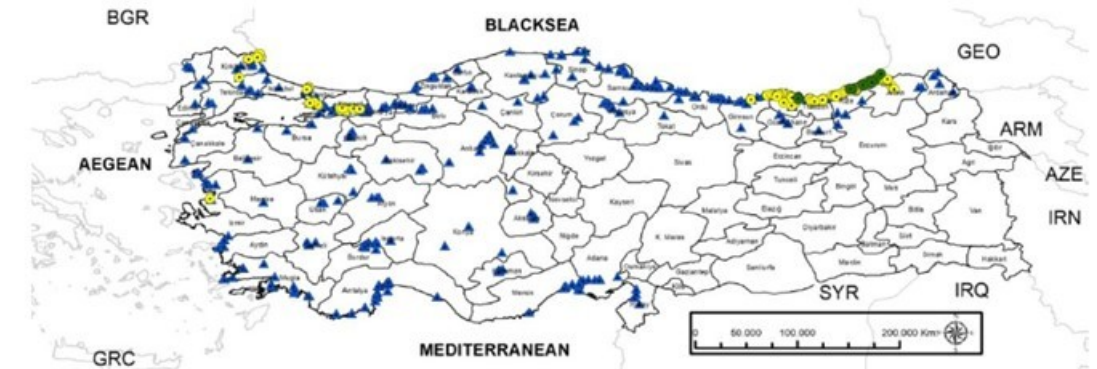


Figure 7. Distribution of *Aedes albopictus* and *Aedes aegypti* in northeastern Turkey in 2019. Blue triangles; monitored sampling sites, yellow circles; positive sampling sites for *Ae. albopictus* and green circles; positive sampling sites for *Ae. aegypti*.

RESEARCH ARTICLE

Arboviral screening of invasive *Aedes* species in northeastern Turkey: West Nile virus circulation and detection of insect-only viruses

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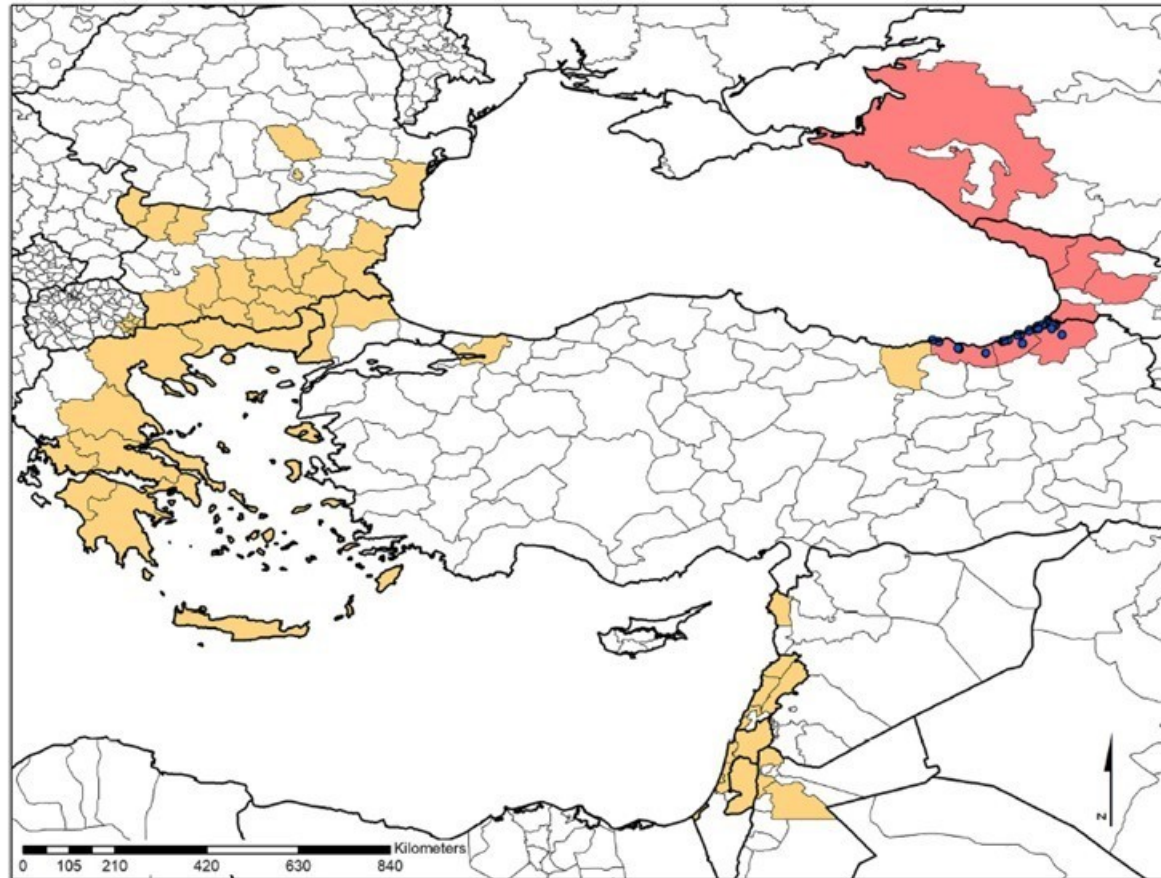
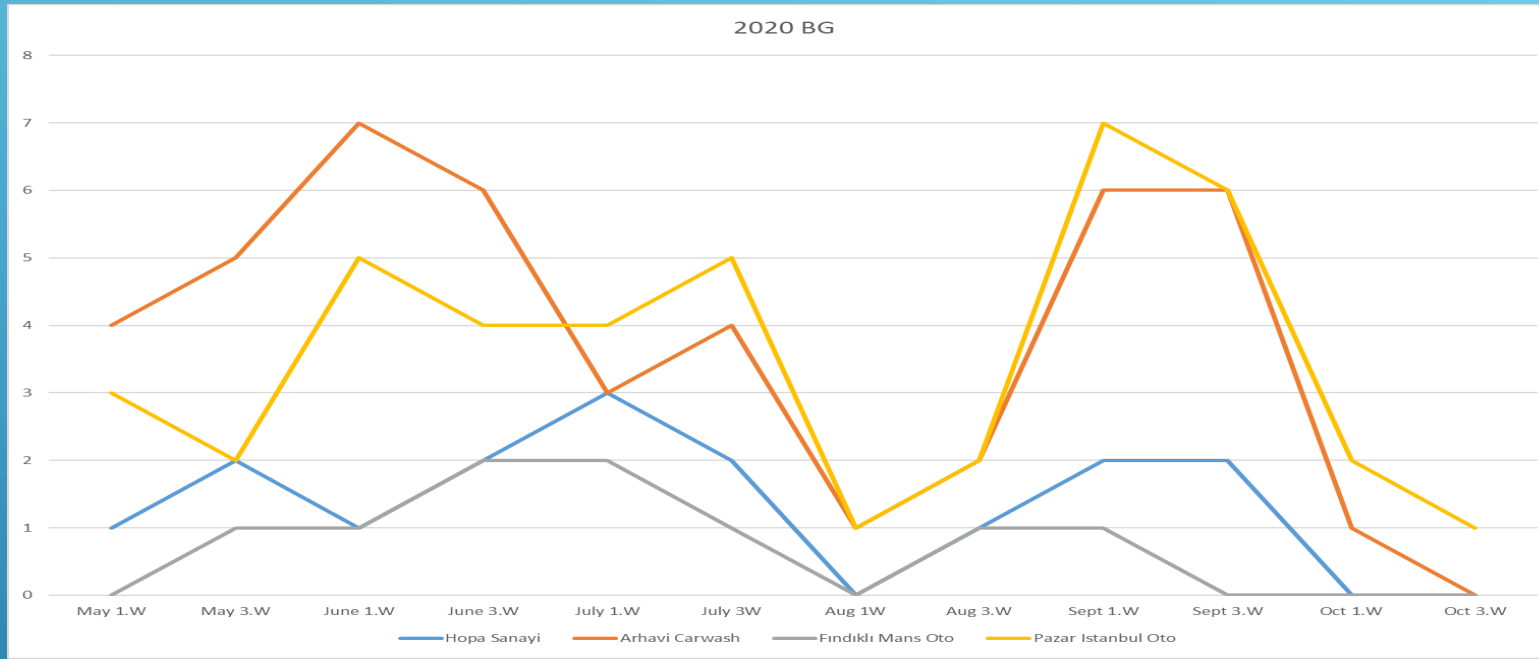


Fig 1. Map of the locations used for mosquito collection in the study. (Red: countries/territories with *Ae. aegypti* and *Ae. albopictus*; Orange: countries/territories with *Ae. albopictus*). Blue dots represent sampling locations. The baseline map has been prepared using Natural Earth raster + vector map data in the public domain (URL: www.naturalearthdata.com, Accessed: April 2019), which is freely available for personal, educational, and commercial use. Current information on *Aedes* species were obtained from the European Centre for Disease Prevention and Control websites (<https://ecdc.europa.eu/en/publications-data/aedes-egypti-current-known-distribution-june-2018>; <https://ecdc.europa.eu/en/publications-data/aedes-albopictus-current-known-distribution-june-2018>; Accessed: December 2018).

Table 2. The mosquito pools with detectable virus sequences.

	Pool Code	Province	Site	Species-Pool content	Virus detected	Sequence characterized
1	KRD15	Artvin	A10	<i>Ae. albopictus</i> (21♀)	CFAV	Partial NS5 (MF361262)
2	KRD22	Artvin	A5	<i>Ae. albopictus</i> (10♀)	CFAV	Partial NS5 (MF361264)
3	KRD24	Rize	R1	<i>Ae. aegypti</i> (12♀)	CFAV	Partial NS5 (MF361265)
4	KRD30	Rize	R2	<i>Ae. aegypti</i> (10♀)	CFAV	Partial NS5 (MF361263)
5	KRD1	Artvin	A3	<i>Ae. albopictus</i> (13♀)	AEFV	Partial NS5 (MF361267)
6	KRD5	Rize	R2	<i>Ae. aegypti</i> (12♀)	AEFV	Partial NS5 (MF361268)
7	KRD32	Artvin	A3	<i>Ae. albopictus</i> (11♀)	AEFV	Near-complete genome (MK251047)
8	11	Artvin	A13	<i>Ae. albopictus</i> (4♀)	AEFV	Partial NS5 (MK251048)
9	20	Rize	R1	<i>Ae. albopictus</i> (21♀)	AEFV	Partial NS5 (MK251049)
10	26	Artvin	A15	<i>Ae. albopictus</i> (14♀)	AEFV	Partial NS5 (MK251050)
11	6	Artvin	A7	<i>Ae. albopictus</i> (2♀)	WNV	Partial NS5 (MK251051)
12	10	Artvin	A11	<i>Ae. albopictus</i> (33♀)	WNV	Partial NS5 (MK251052)
13	21	Rize	R6	<i>Ae. albopictus</i> (1♀)	WNV	Partial NS5 (MK251053)
14	24	Artvin	A15	<i>Ae. albopictus</i> (7♀)	WNV	Partial NS5 (MK251054)
15	17	Artvin	A1	<i>Cx. pipiens</i> s.s. (5♀)	WNV	Partial NS5 (MK251055)
					MERLVT	L region (MK251056)
						N region (S3 Fig)



Checked four points routinely with Bg sentinel with lure Hopa

2020

Arhavi

Fındıklı

Pazar

ANCIENT REMNANT OR NEW INVASION

- Kotsakiozi et al. (2018). *Aedes aegypti* in the Black Sea: recent introduction or ancient remnant? *Parasites and Vectors* 11 (1): 1-13. implied that Black Sea populations being remnants of older populations, likely the now extinct Mediterranean populations that, consistent with the historic epidemiological record, likely represent the original return of *Ae. aegypti* to the Old World.
- Our different two study
- One with four mitochondrial and one genomic DNA markers. our results revealed that our region (Turkey and Georgia) samples were most probably a new invasion rather than an ancient one (Ozturk and Akiner (2023) "Molecular phylogenetics of *Aedes aegypti* (L., 1762) (Diptera: Culicidae) in Eastern Black Sea area of Turkey and possible relations with the Caucasian invasion," *Turkish Journal of Zoology*: Vol. 47: No. 3, Article 4.
- The other one we studied resistance profiles. The resistance profiles observed in *Ae. aegypti* populations may be partially shaped by the origin and introduction history of the invasive lineage. As the species is likely of Asian origin, it is possible that the founding populations introduced into the region had not previously been exposed to strong insecticide selection pressures, or that resistance levels reflect genetic bottlenecks associated with the colonization process. (Ozturk et al 2025 'Pyrethroid and Organochlorine Susceptibility in *Aedes aegypti* Populations from Eastern Black Sea Region. Türkiye: Traces in the *kdr* Gene Region of the Species Origin' accepted but not published yet)

- ▮ Surveillance activity conducted two spot point
- ▮ Hopa, Pazar during active season with BG sentinel and Ovitrap
- ▮ Ovitrap results resolution not consistent and very complicated related to the mix of *Aedes albopictus*, therefore we use BG sentinel data

2021-2022-2023



Pazar pilot test site



Hopa control site

Trap design and locations were produced with QGIS program
We set up traps in the field in May related to the climatic reason
Collect the baseline data during 5 months (May- September)

8.10.2021	COLLECTED	MA	7.10.2021	BG	Pazar	41	10	1	0	1	MA	MA	0	1	
8.10.2021	COLLECTED	MA	7.10.2021	BG	Pazar	41	10	1	0	1	MA	MA	0	1	
8.10.2021	COLLECTED	MA	7.10.2021	BG	Pazar	41	10	0	0	0	MA	MA	0	0	
8.10.2021	COLLECTED	MA	7.10.2021	BG	Pazar	41	10	2	1	1	MA	MA	1	1	
8.10.2021	COLLECTED	MA	7.10.2021	BG	Pazar	41	10	0	0	2	MA	MA	0	2	
8.10.2021	COLLECTED	MA	7.10.2021	BG	Pazar	41	10	0	0	0	MA	MA	0	0	
8.10.2021	COLLECTED	MA	7.10.2021	BG	Pazar	41	10	0	0	0	MA	MA	0	0	
DateColect	Trap status	responsibl	Date0	rap_syster	Site	Week	Month	egypti Adu	Aegypti males	gypti fema	FieldInsp	LabInsp	males/trap	males/trap/day	

2021

2022

2023

Satr Etiketleri	Toplam Aegypti males	Toplam Aegypti females
20	0	0
21	0	0
22	3	0
23	21	45
24	34	51
25	74	52
26	114	92
27	90	48
29	54	45
30	33	26
31	9	34
32	2	11
33	2	20
34	15	20
35	23	16
36	22	24
37	21	43
38	28	54
39	12	49
40	8	34
41	2	18
Genel Toplam	567	682

Satr Etiketleri	Toplam Aegypti Adults	Toplam Aegypti males	Toplam Aegypti females
19	0	0	0
20	0	0	0
21	2	2	0
22	13	2	11
23	35	1	34
24	5	1	4
25	31	9	22
26	27	14	13
27	30	9	21
28	30	25	5
30	4	2	2
33	14	2	12
34	18	15	3
35	26	23	3
36	49	43	6
37	26	25	1
38	17	1	16
39	10	4	6
40	2	2	0
41	1	1	0
Genel Toplam	340	181	159

Satr Etiketleri	Toplam Aegypti Adults	Toplam Aegypti males	Toplam Aegypti females
19	0	0	0
20	0	0	0
21	0	0	0
22	1	0	1
23	5	3	2
24	0	0	0
25	3	0	3
26	2	2	0
27	3	1	2
28	2	1	1
29	3	1	2
30	4	2	2
31	15	3	12
32	17	11	6
33	6	2	4
34	0	0	0
36	0	0	0
37	1	0	1
38	2	1	1
39	0	0	0
40	0	0	0
41	0	0	0
Genel Toplam	64	27	37

- ▯ During last Five years,
- ▯ Population decreased step by step
 - ▯ Possible reason
 1. *Ae. albopictus* invasion and spread every where very quickly and supress
 2. Possible climate effects,
 - ▯ Reinfall regime and very heat season
 3. Very sensitivity to the insecticides (Ozturk et al. 2025)

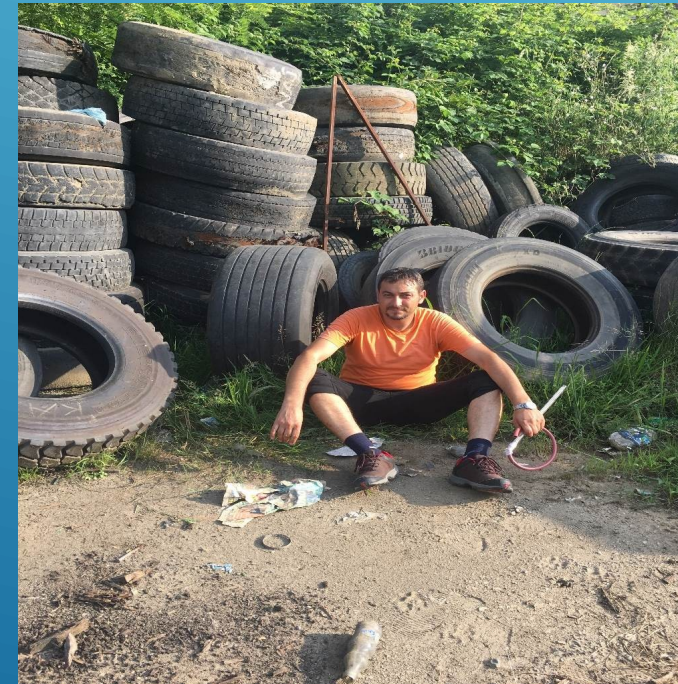
AS A RESULT

MOSQUITO CONTROL PROGRAM ACTIVITIES FROM 2021-2023

- ▮ Just only municipalities conduct the activities but we have no data about the control activities
- ▮ Generally used the bti and bs for larval control related to the rules in the city centers but not more
- ▮ Pyrethroid based adult control activities
- ▮ Not collected baseline data
- ▮ Just only preferred individual application of the insecticides in the area

THE MAIN CHALLENGES AND NEEDS OF TÜRKİYE

- ▮ Lack of coordination between the control units and Universities
- ▮ Continuing old habits in the areas in specifically Blacksea part
- ▮ Lack of training of control staff for invasive species







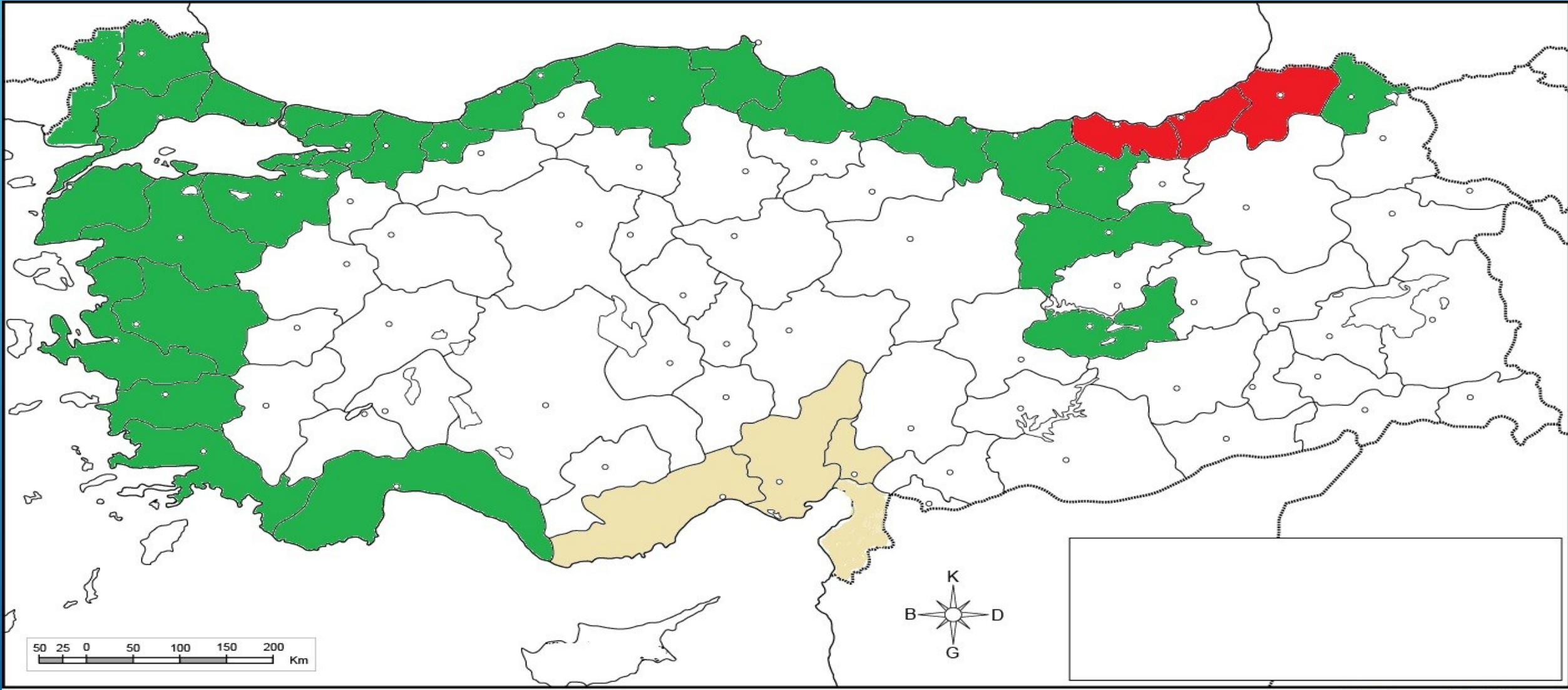




**PAZAR PİLOT SITE AREA OVİTRAP AND HLC AND BG
COLLECTION NEAR THE HOUSE**
Tomb: Natural breeding habitat near the house in Hopa

- ▮ Not for aegypti and also albopictus
- ▮ Community participation Key factor for old habits changing and eradication of the breeding points
- ▮ Coordination of the control units (especially Municipality)
- ▮ Use all options for eradication and cut of reestablish and increase population levels.
- ▮ Biological, Chemical, Wolbacchia, SIT ??????

WHAT WE DO?



Last confirmed situation in 2024

□ Thanks for listening

Special thanks

My Students

Yusuf Emre Doğan, MSc Eyüphan Çekiç, Msc Sümeyye Gündüz

PhD Murat Ozturk

From Turkish Ministry of Health Turkish Ministry of Health General
Directorate of Public Health Zoonotic and Vectorial Diseases Department

Dr. Seher Topluoğlu, Msc Ünal Altug, MSc Ahmet Ferhat Doğan

Three parallel white lines of varying lengths and slopes are positioned in the bottom right corner of the slide, extending from the right edge towards the center.